UAiTED
University Alliance in Talent Education Development

6 CLEAN WATER AND SANITATION
7 AFFORDABLE AND CLEAN ENERGY
9 INDUSTRY, INNOVATION AND INFRASTRUCTURE
11 SUSTAINABLE CITIES AND COMMUNITIES

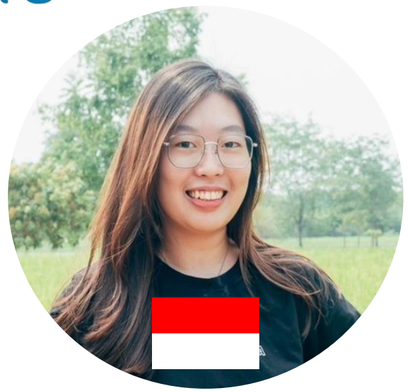
SEASIDERS

Sustainable Water Solutions Consultancy



SEASIDERS

TEAM MEMBERS



Natalie Tanjaya

CEO

Biomedical Engineering
NCKU Taiwan

- Experienced in electrical and technological innovation researches, including wearable devices, AI for medical applications, and robotic control.
- Top 8 Finalists at Hult Prize Mumbai Summit 2024

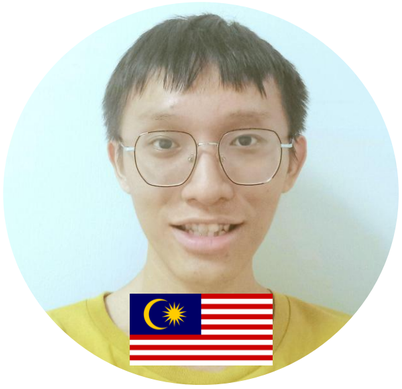


Chai Jin Xie

CTO

Civil Engineering
NTU Singapore

-Developed business models for multiple startup related competitions
-Silver Prize in NTU Port63 Challenge and Bronze Price in UAiTED Innovation Competition.



Si Poh Yi

CMO

Civil Engineering,
Urban Systems
NTU Singapore

- Familiar with market research and global current affairs.
- Chairperson of international event that attracted record highest participants through marketing skills.



Lam Nguyen Ngan Ha

CFO

Biomedical Engineering
NCKU Taiwan

- Research interests in IoT wearable device, robotic control, women and fetus health care, AI application on processing


Do You Know Where is This ???


Do You Know Where is
This???


Do You Know Where is
This???

This is TAIWAN !

Before Drought







After Drought



Wushe Reservoir in Nantou County



Sun Moon Lake in Nantou County



Ai Liao river bed in Pingtung county

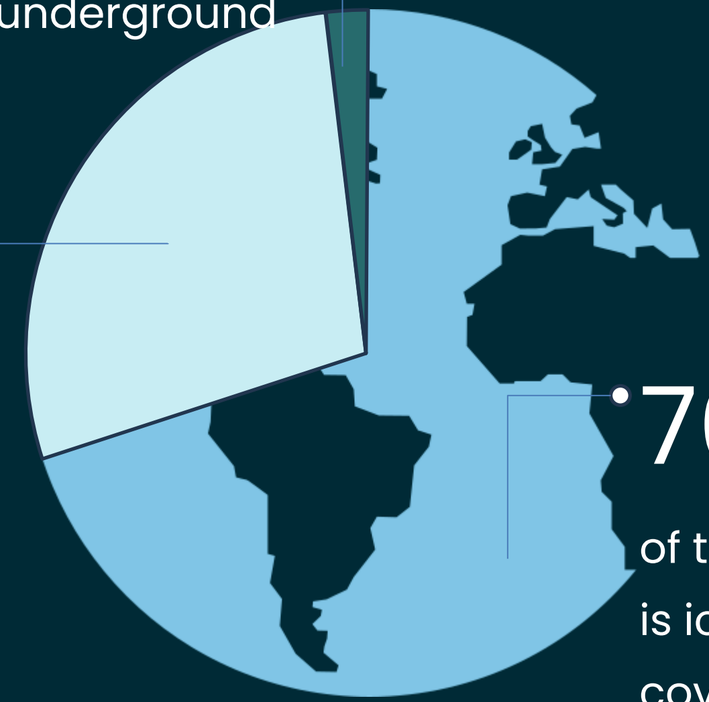
We Need Water !

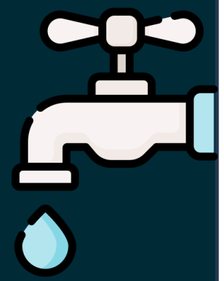
- **29%**

of the world’s freshwater is underground

- **1%**

of the world’s freshwater is easily accessible for **human use**

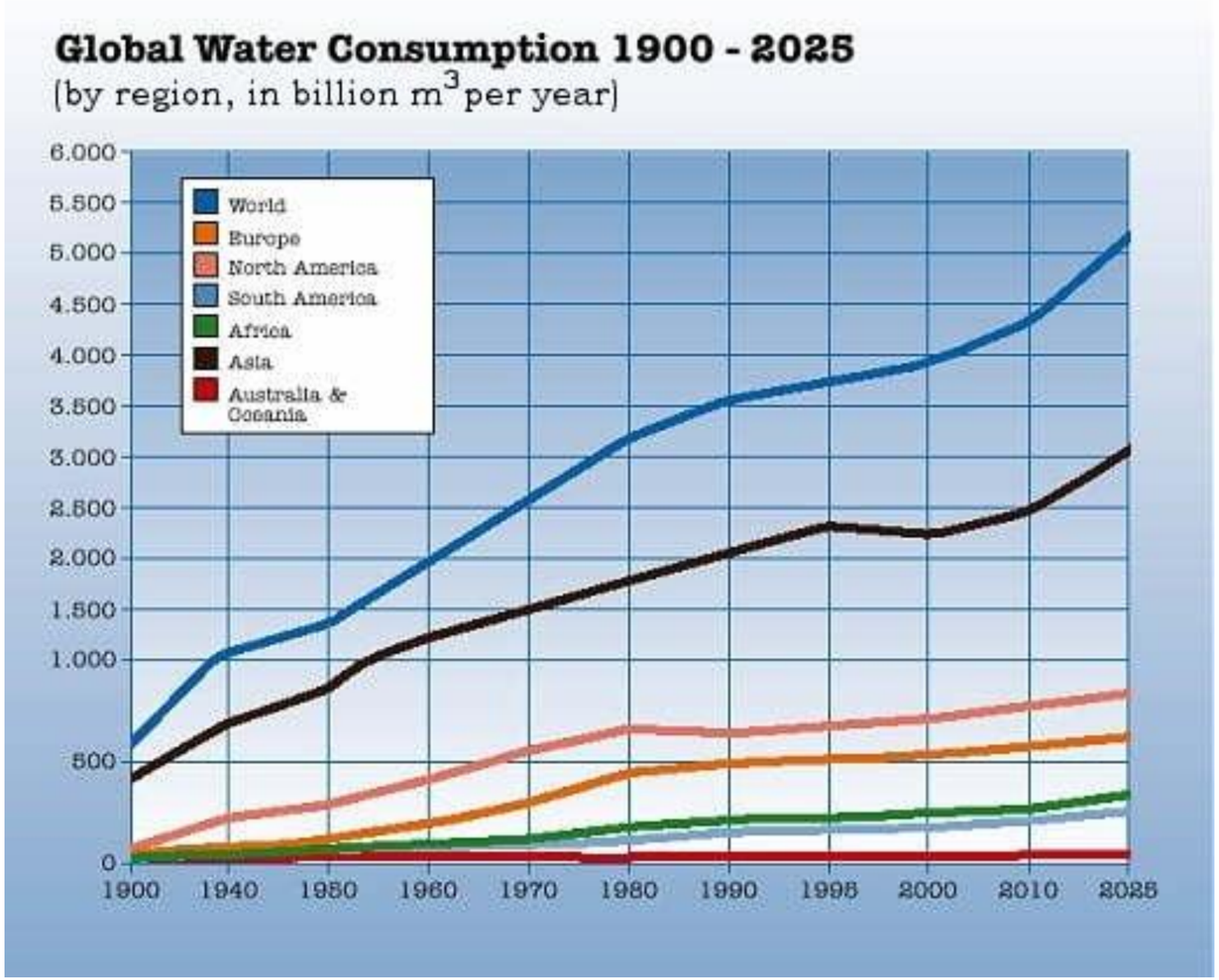




- **70%**

of the world’s freshwater is ice form or snow cover on mountains

Source: National Geographic, 2024



Source: Global Water Report, 2024

Global Water Crisis

- **Taiwan**'s reservoir goes down to 3 months' water supply

  (Source: Water Resources Agency, 2024)

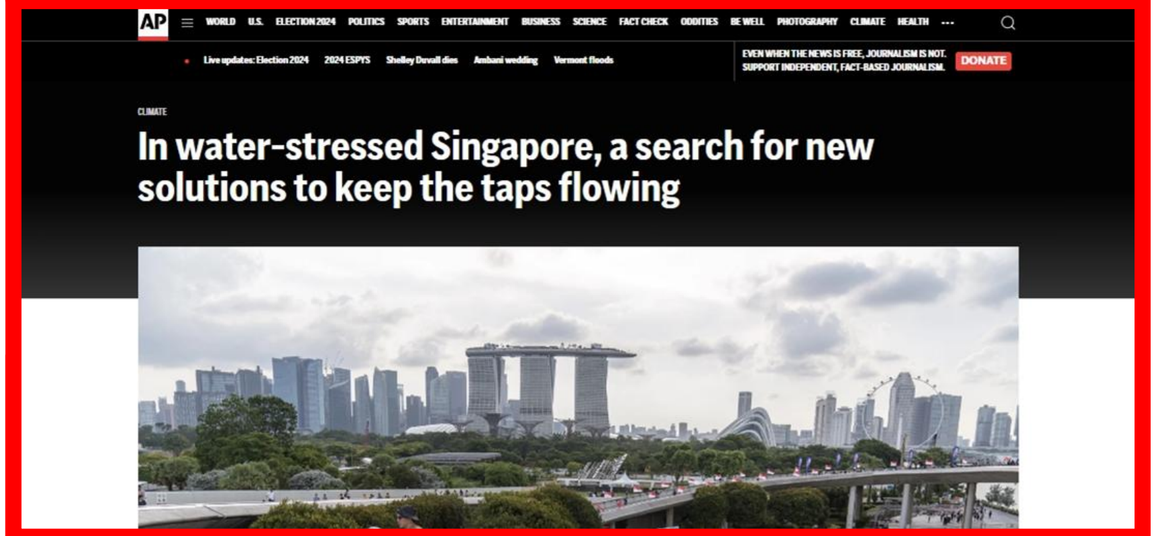
- **Singapore** is amongst the most water-stressed countries in the world

  (Source: MSE Singapore, 2024)

- **Saudi Arabia** is a desert country with no permanent rivers or lakes and very little rainfall.

  (Source: The Embassy of the Kingdom of Saudi Arabia, 2024)

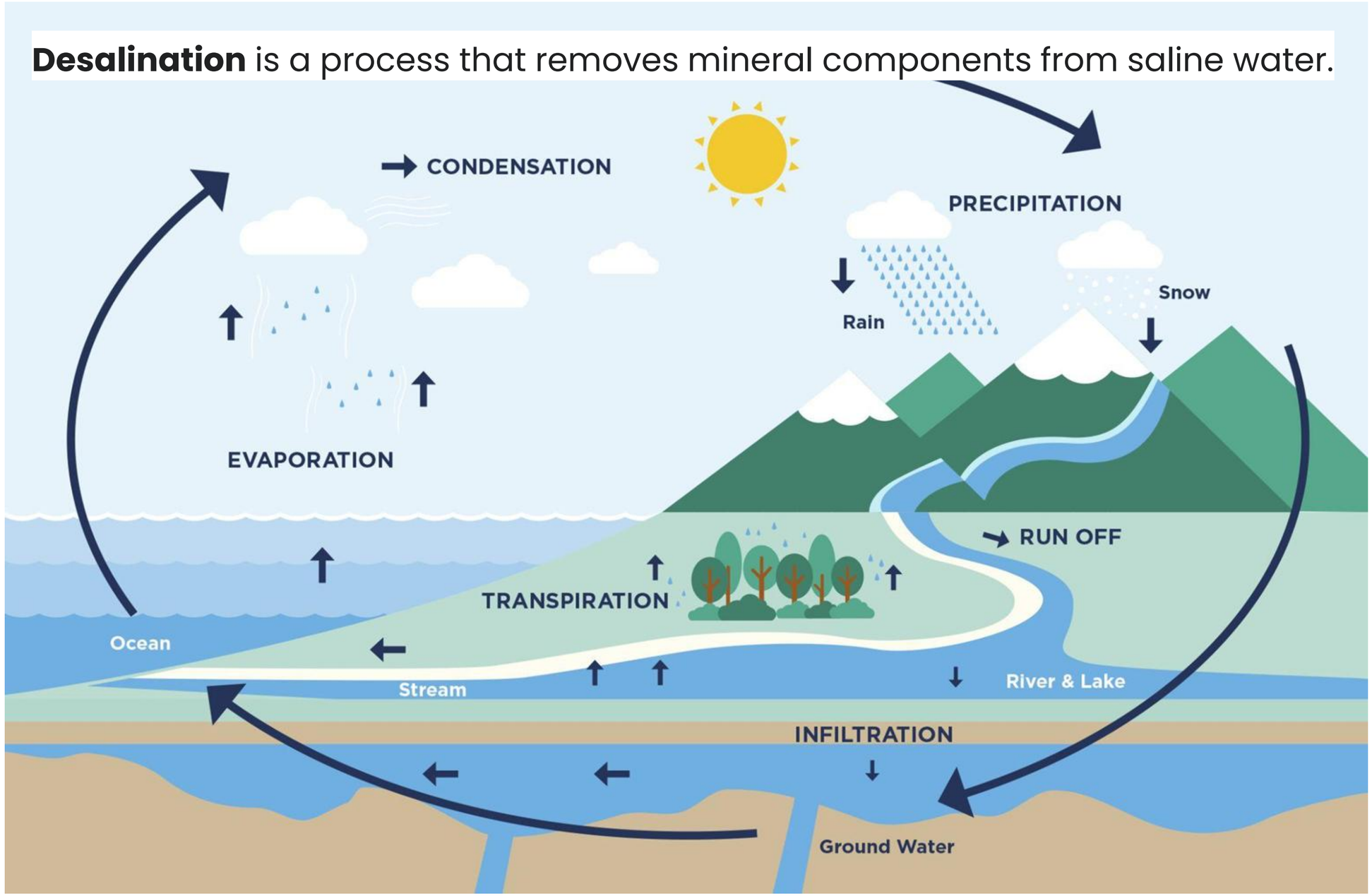






WHAT IS DESALINATION ?

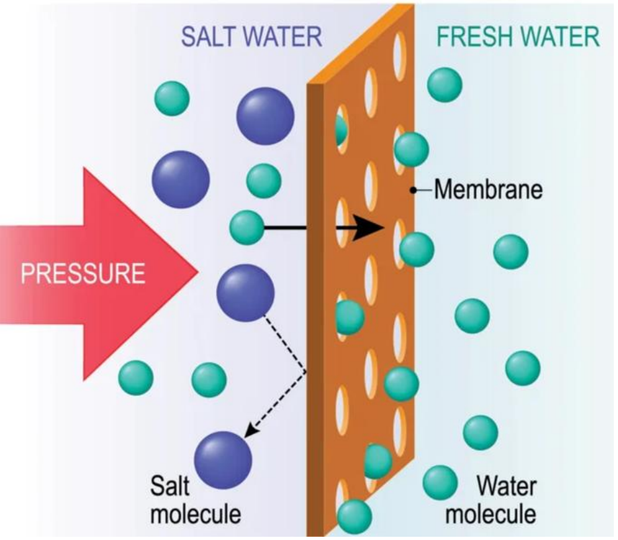
Desalination is a process that removes mineral components from saline water.



https://sydneydesal.com.au/education/desalination-and-the-water-cycle/

EXISTING SOLUTIONS

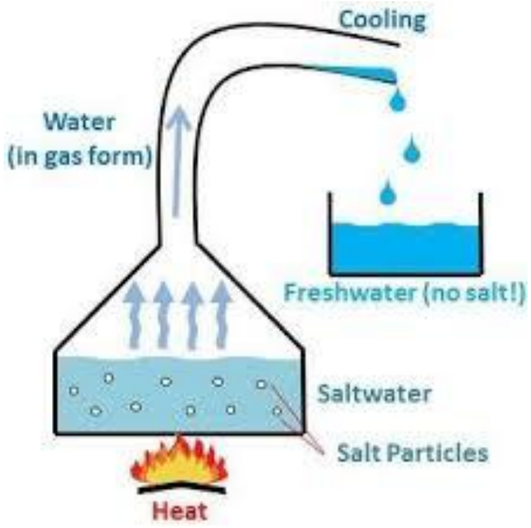
Reverse Osmosis



Membrane fouling

3 – 5 KWh/m³

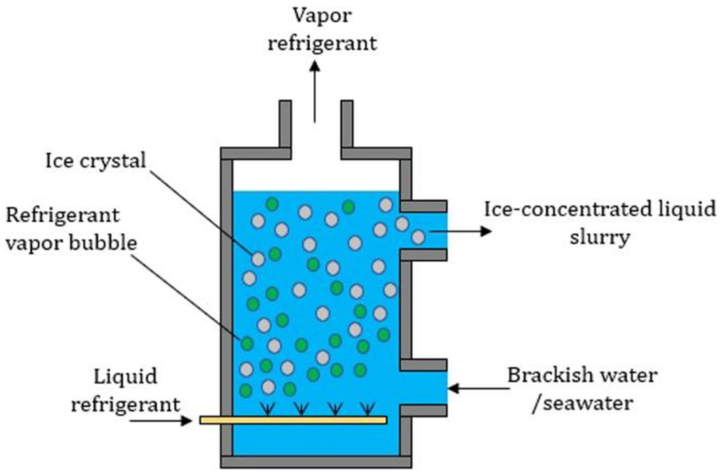
Thermal Distillation



Dependence on fossil fuels

12 - 25 kWh/m3

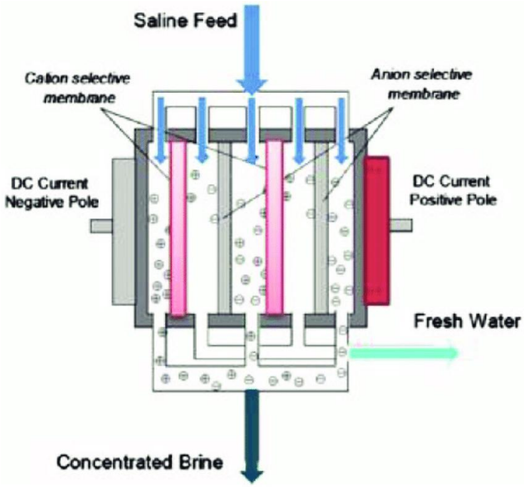
Freeze Desalination



High complexity

50 - 90 kWh/m3

Electrodialysis



Seawater is not feasible

1 - 4 kWh/m3

Pain Points of Current Desalination Plants



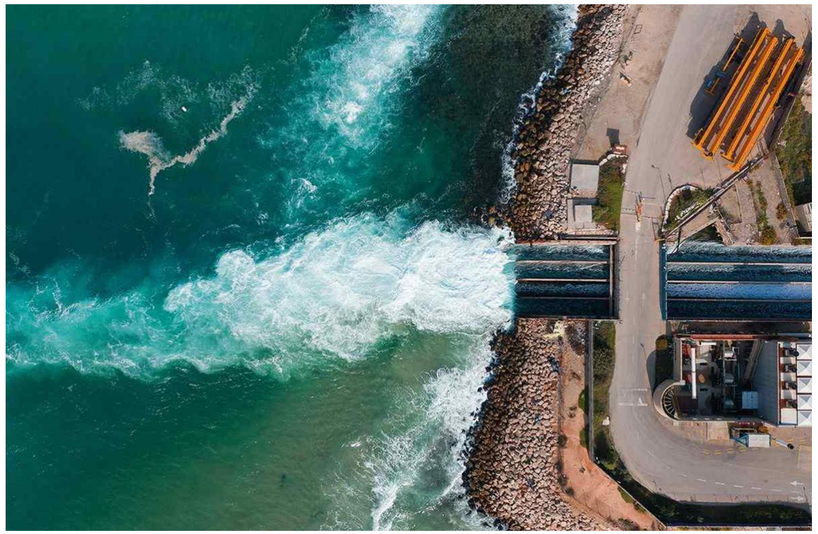
Emergency seawater desalination plant in Taichung, Taiwan



The large-scale, dual-mode Keppel Marina East Desalination Plant in Singapore



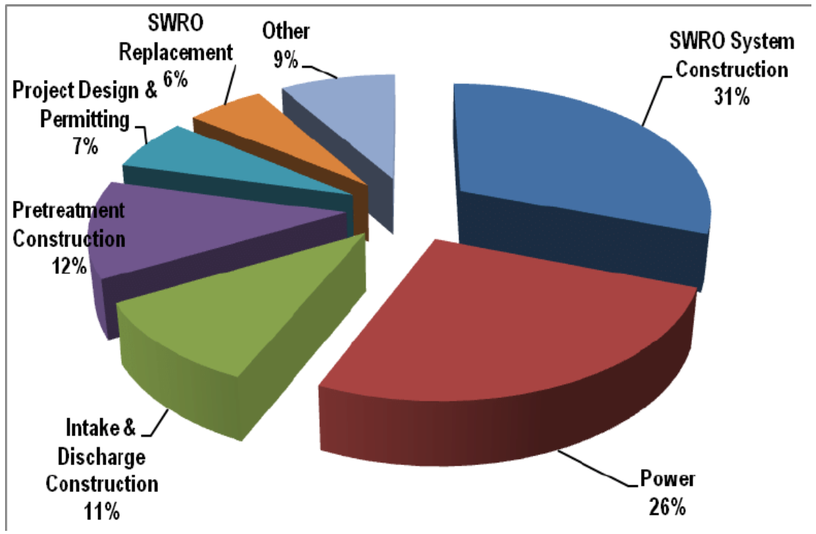
New Al Jubail SWRO Desalination Plant in the Easter Region of Saudi Arabia



Not eco-friendly



Energy intensive



High cost



SEASIDERS

Sustainable Water Solutions Consultancy

Lateral Thinking with Withered Technology

(Forward Omosis)

FO

- Wastewater treatm
- Low to high
- Natural process

(Pressure Retarded Omosis)

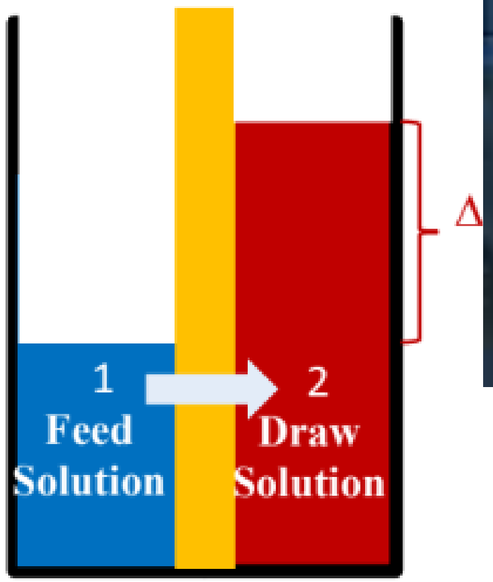
PRO



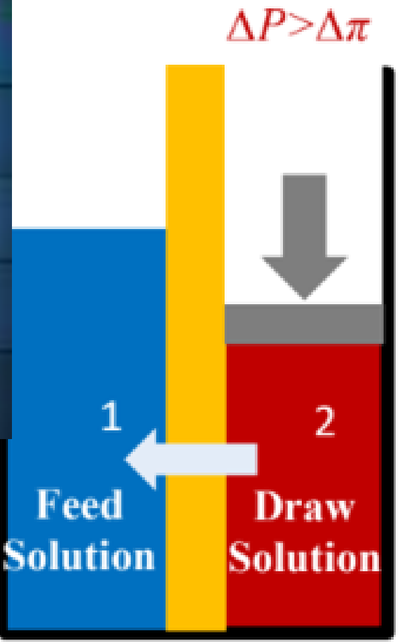
(Reverse Omosis)

RO

- Desalination
- High to Low
- Reject salinity

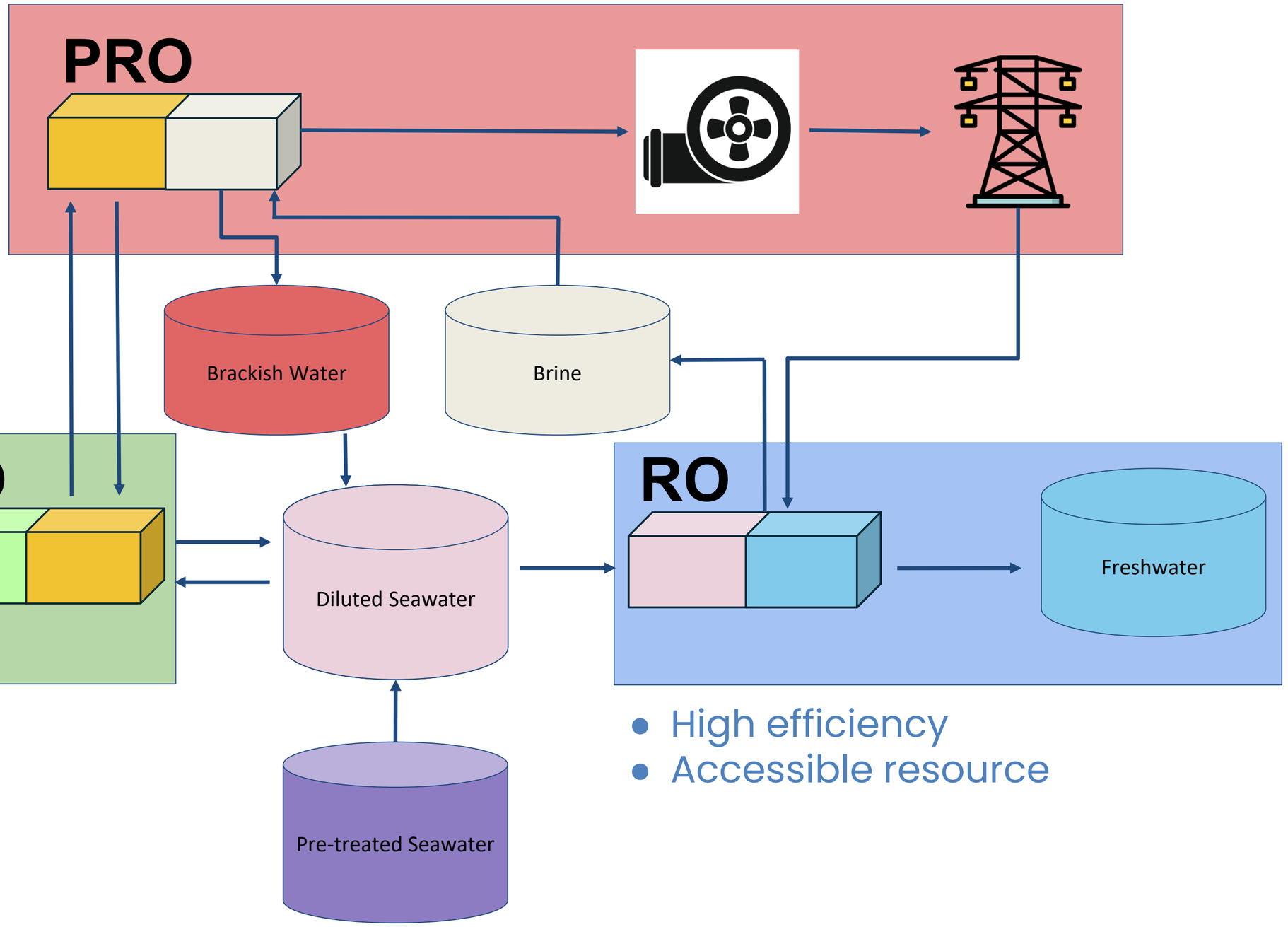






Solution Framework

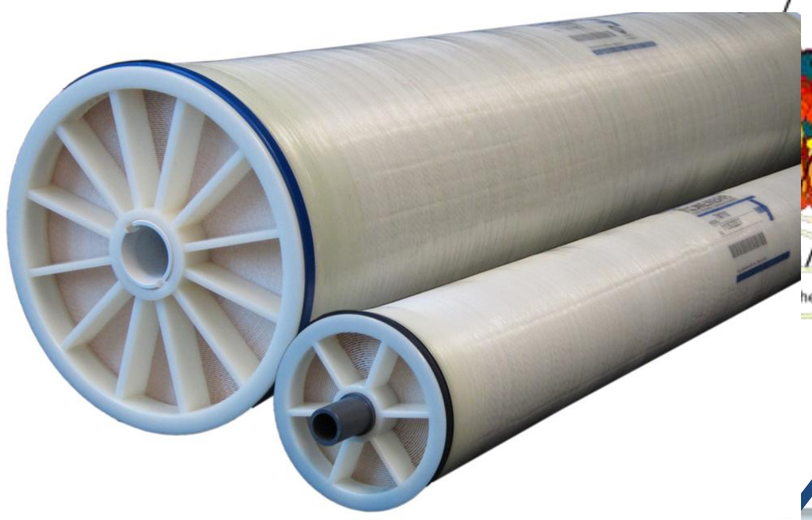
- Minimal environmental impact
- Continuous energy generation

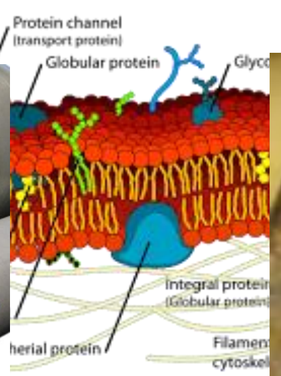


- Salinity control
- Reduce membrane fouling risks

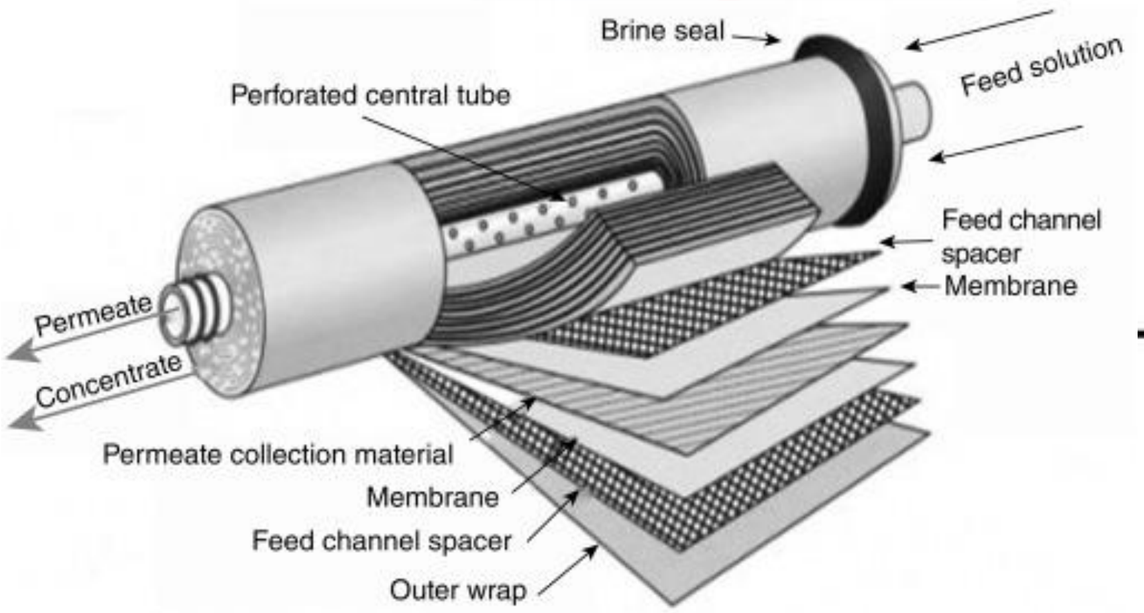
- High efficiency
- Accessible resource

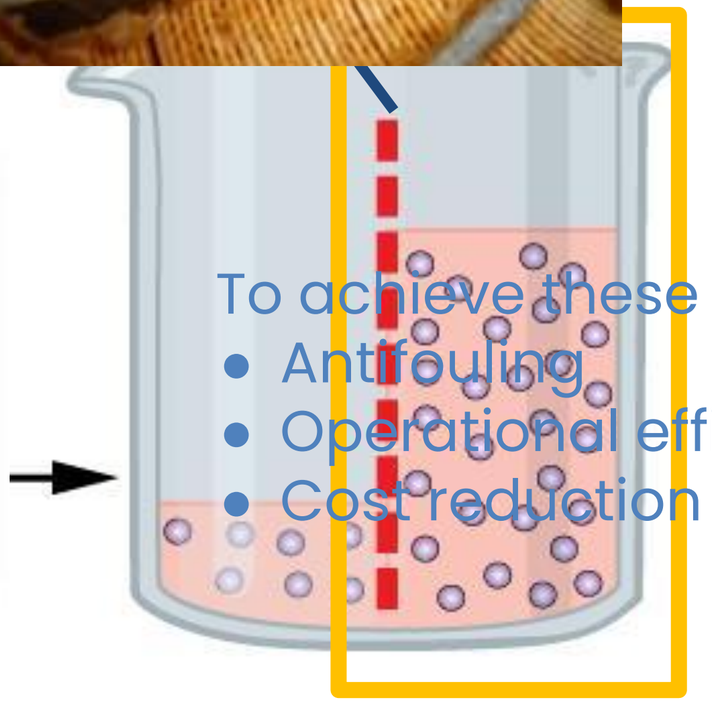
KEY COMPONENT – MEMBRANE TECHNOLOGY









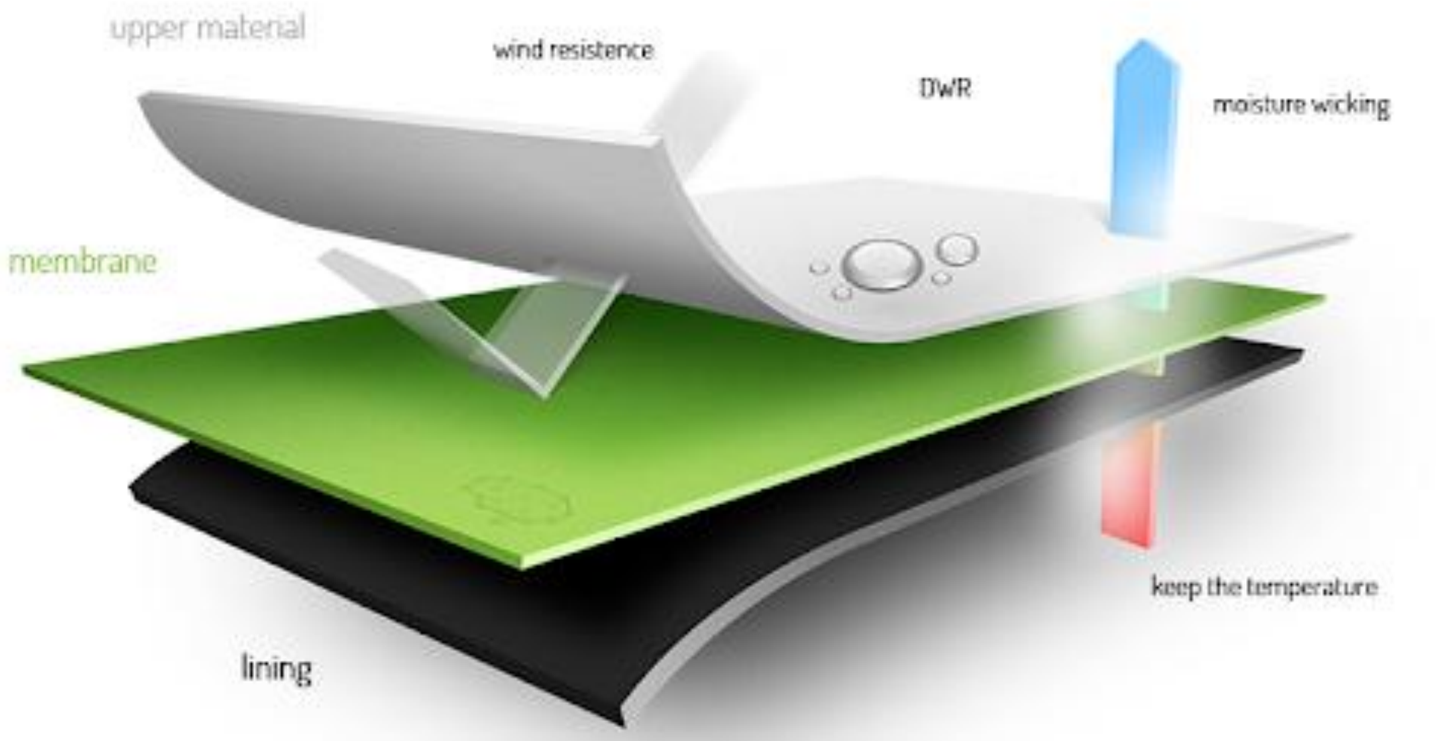


To achieve these objectives

- Antifouling
- Operational efficiency
- Cost reduction

Pressure increase

MATERIAL BREAKTHROUGH - NANOMEMBRANE

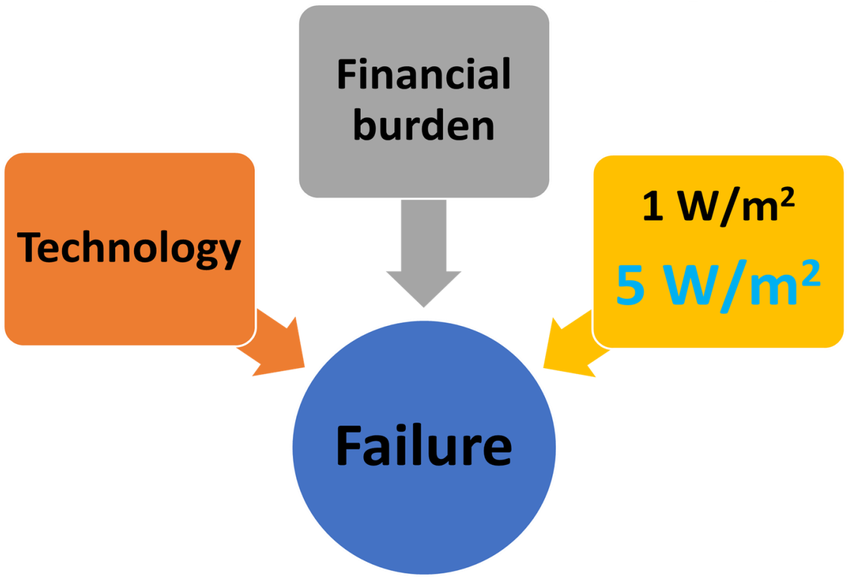


- Enhanced filtration
- Anti-fouling
- High stability
- Customizable

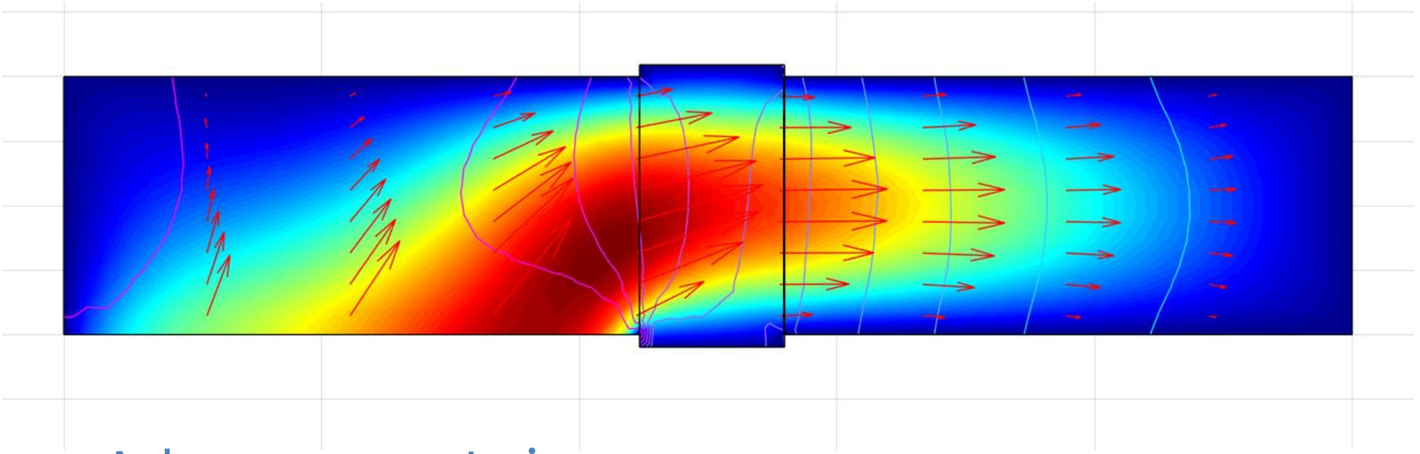
CURRENT LANDSCAPE OF OSMOTIC ENERGY

2014: Statkraft discontinues investment





2024: Promising opportunity



Advancements in:

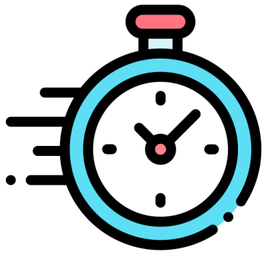
- Material Engineering
- Artificial Intelligence (AI)
- Computational Fluid Dynamics (CFD)

Improvements in following:

- Efficiency
- Cost-effectiveness

https://www.forwardosmosistech.com/statkraft-discontinues-investments-in-pressure-retarded-osmosis-2/

Solution Impacts



Faster

- Short construction time
- Quick deployment



Reliable

- Consistency
- Less vulnerable



Better

- Energy efficiency
- Resource recovery



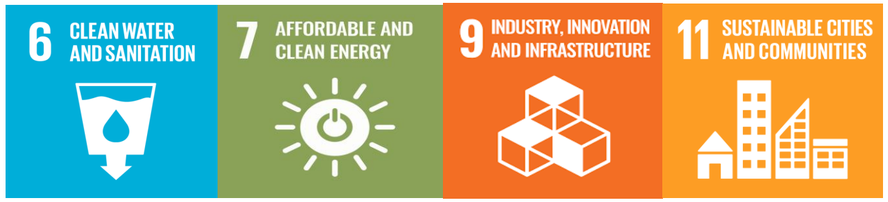
Sustainable

- Clean energy
- Minimal environment impact



Cheaper

- Lower costs
- Scale down size



OUR BUSINESS SERVICES

Strategic Planning



- Site Selection
- Feasibility Studies
- Regulatory Compliance
- Sustainability Integration

Implementation Design & Maintenance



- Architecture Design
- Technology Selection
- Project Management
- Risk Mitigation
- Maintenance & QC

Advanced Analytics

Balance Sheet (€)	Start	Dec-2023	Dec-2024	Dec-2025
ASSETS				
Tangible assets	301,889	271,700	241,511	211,322
Intangible assets	0	0	0	0
Financial assets	0	0	0	0
Total fixed assets	**301,889**	**271,700**	**241,511**	**211,322**
Inventory	0	4,478	4,657	4,843
Account receivables	17,711	21,226	22,075	22,958
Advances paid	0	0	0	0
Deferred costs	0	0	0	0
Net fiscal and social assets	0	0	0	0
Cash	98,199	144,145	223,096	313,237
Total current assets	**115,909**	**169,849**	**249,828**	**341,038**
Other assets	0	0	0	0
TOTAL ASSETS	**417,799**	**441,549**	**491,339**	**552,360**
LIABILITIES & EQUITY				
Account payables	90,567	62,847	65,306	67,862
Advances received	0	0	0	0
Deferred revenues	0	0	0	0
Net fiscal and social debts	0	11,679	13,202	15,702
Financial debt	25,000	19,210	13,123	6,725
Overdraft	0	0	0	0
Accrued interests	0	0	0	0
Total debts	**115,567**	**93,736**	**91,632**	**90,289**
Capital	100,630	100,630	100,630	100,630
Shareholders' loan	0	0	0	0
Reserves	0	0	0	0
Investment grants	0	0	0	0
Retained earnings	201,602	247,184	298,078	361,442
Other equity	0	0	0	0
Total Equity	**302,232**	**347,814**	**399,708**	**462,071**
Provisions	0	0	0	0
Other liabilities	0	0	0	0
TOTAL LIAB. & EQUITY	**417,799**	**441,549**	**491,339**	**552,360**

- Financial Modeling
- AI Simulations
- Data-driven Optimization
- Performance Monitoring

PROJECT TIMELINE

Consultant's Role at Each Project

Concept and Feasibility stage

- Conduct feasibility of technical and economic analysis.

Planning and Design stage

- Assist in detailed project plans and engineering design.

Procurement and Contracting Stage

- Develop procurement strategies and contract negotiations.

Construction and Implementation Stage

- Provide management and oversight for logistics and supply chain.

Testing and Commissioning Stage

- Oversee system testing and validation for optimization.

Operational Stage

- Offer technical support and maintenance for performance analysis.

PROJECT TIMELINE

Consultant's Role at Each Stages

1- 3 months	4 - 6 months	7 - 9 months	10 - 15 months	16 - 18 months	18 - 24 months
Concept and Feasibility stage	**Planning and Design stage**	Procurement Contracting Stage	Construction Implementation Stage	Testing Commissioning Stage	Operational Stage
Conduct feasibility of technical and economic analysis.	Assist in detailed project plans and engineering design.	**Assist in detailed project plans and engineering design.**	Provide management and oversight for logistics and supply chain.	Oversee system testing and validation for optimization.	Offer technical support and maintenance for performance analysis.

Our Stakeholders

Government and Regulators

Academic and Research Institutions

Industry Associations

Suppliers (Membrane, Filtration System, Chemical, etc.)

Construction Companies

SEASIDERS

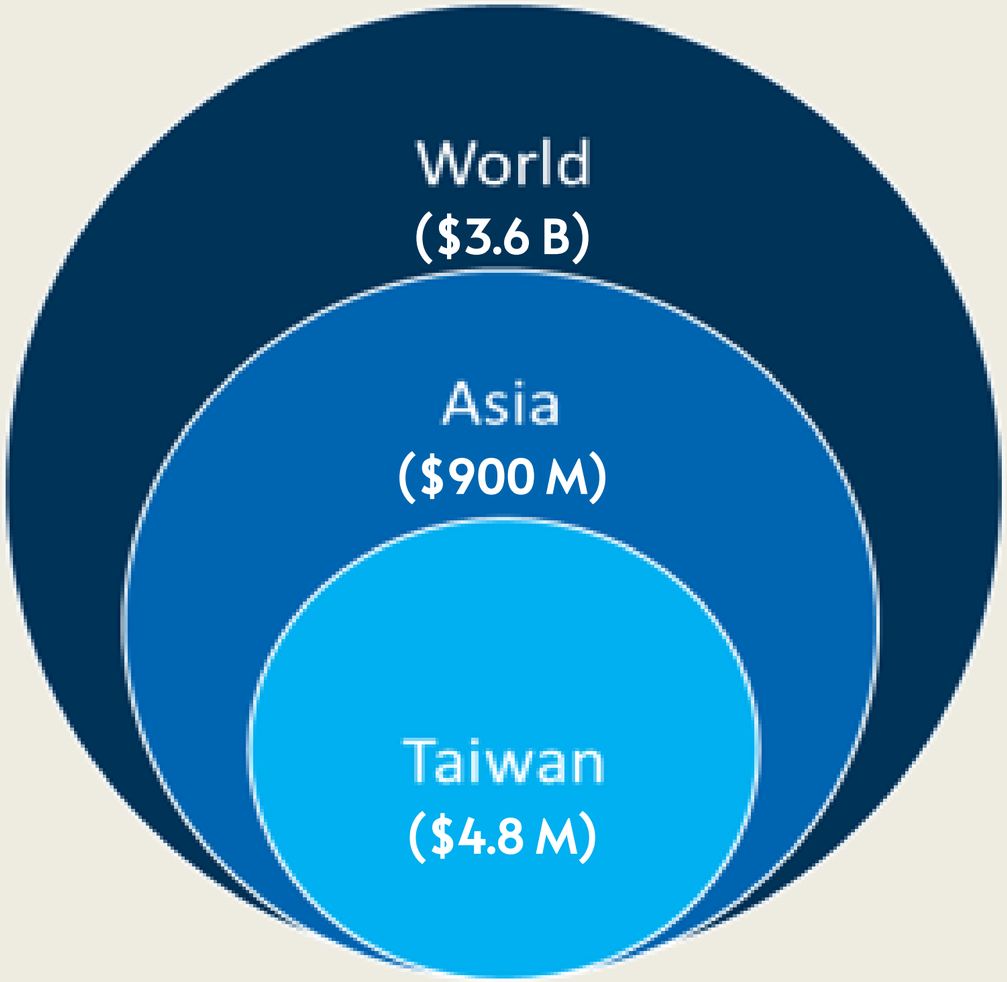
Customers and Stakeholders

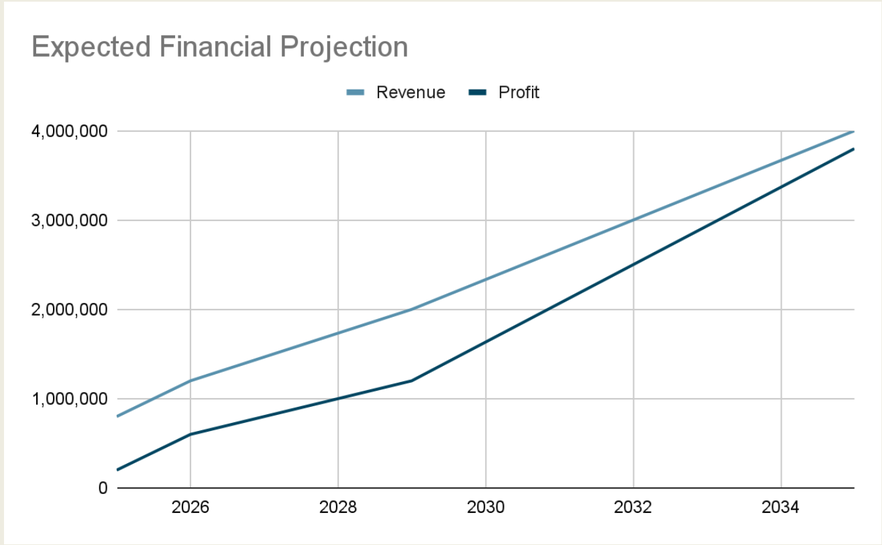
- Municipal Water Supply Company
- Industrial Clients Requiring Desalination
- Agricultural Sector Clients
- Individual and Institutional Investors
- Equity Partners

Strategic Partnerships **Service Provider** **Money Flow**

Market Size & Expected Milestones

(The data below is calculated based on the total number of current existing desalination project in Taiwan, Asia, and Global with estimated consultation fees: US$200,000 per project)





US$ 800,000

First Year Income (Market Test)

Consultation Fees (4) :
US$ 400,000
Project-Based Revenue (1) : US$
200,000
Service Packages (1) :

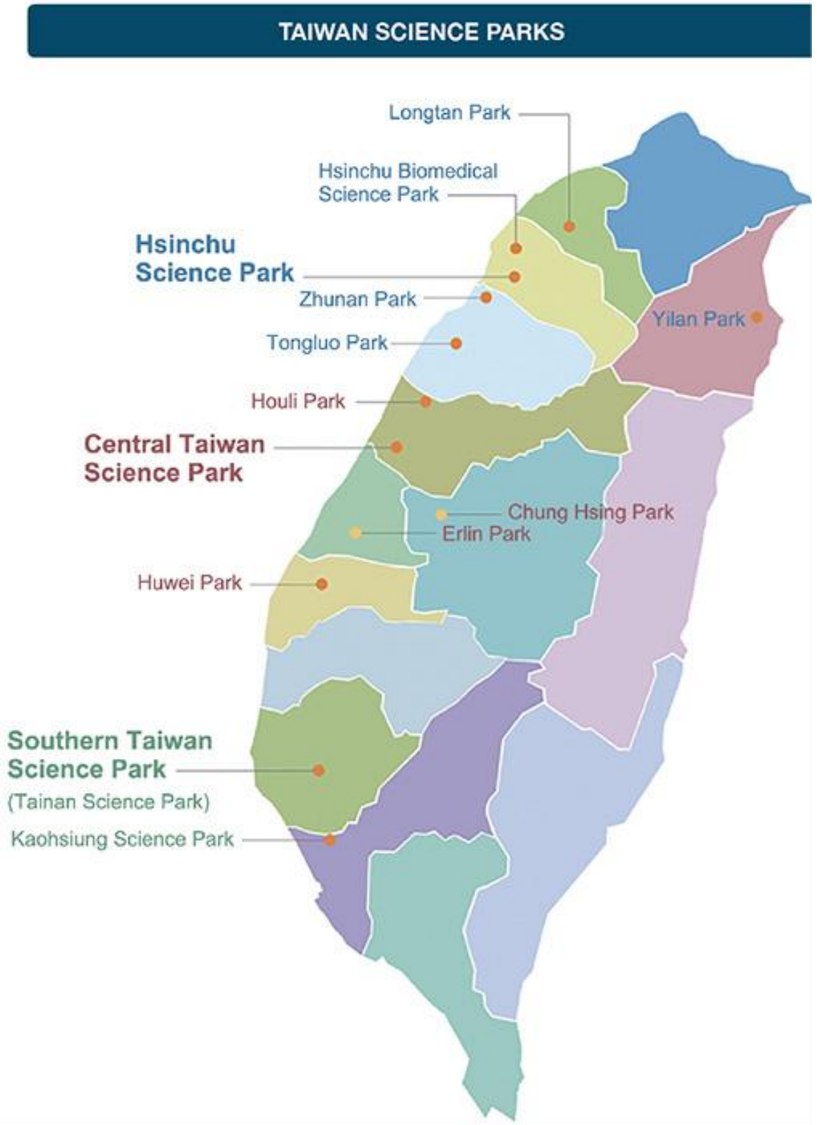
Beachhead Market: Why Taiwan?

CHALLENGES

Droughts & Climate Change

Increasing Water Needs for Industry

Other Water Resources Vulnerable to Rainfall



OPPORTUNITIES

Surrounded by Seawater

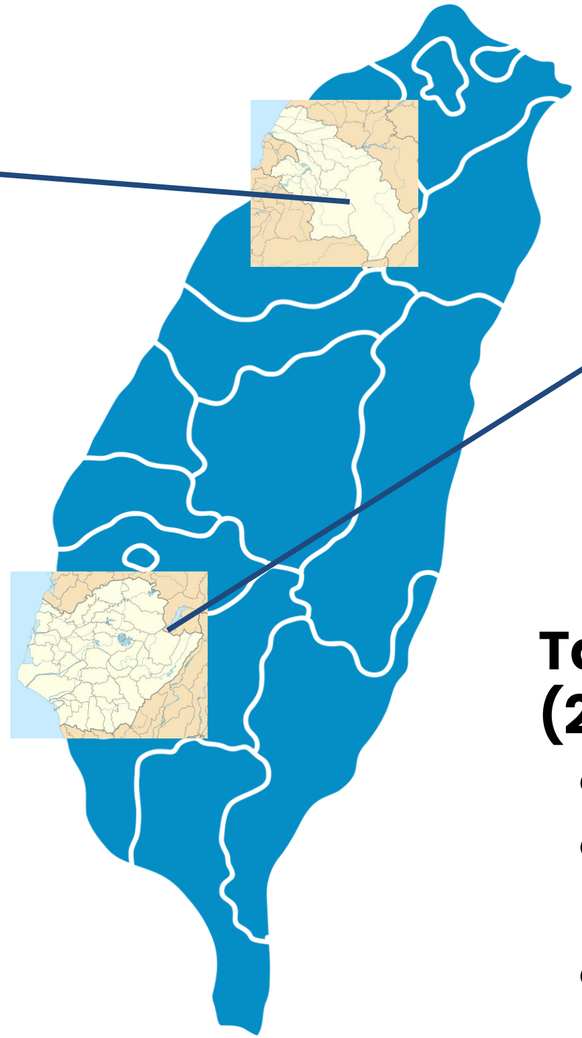
Diversification in Water Supply

Strong Government Support

The Upcoming Plans in Taiwan



Greater Hsinchu Area





Tainan and Kaohsiung

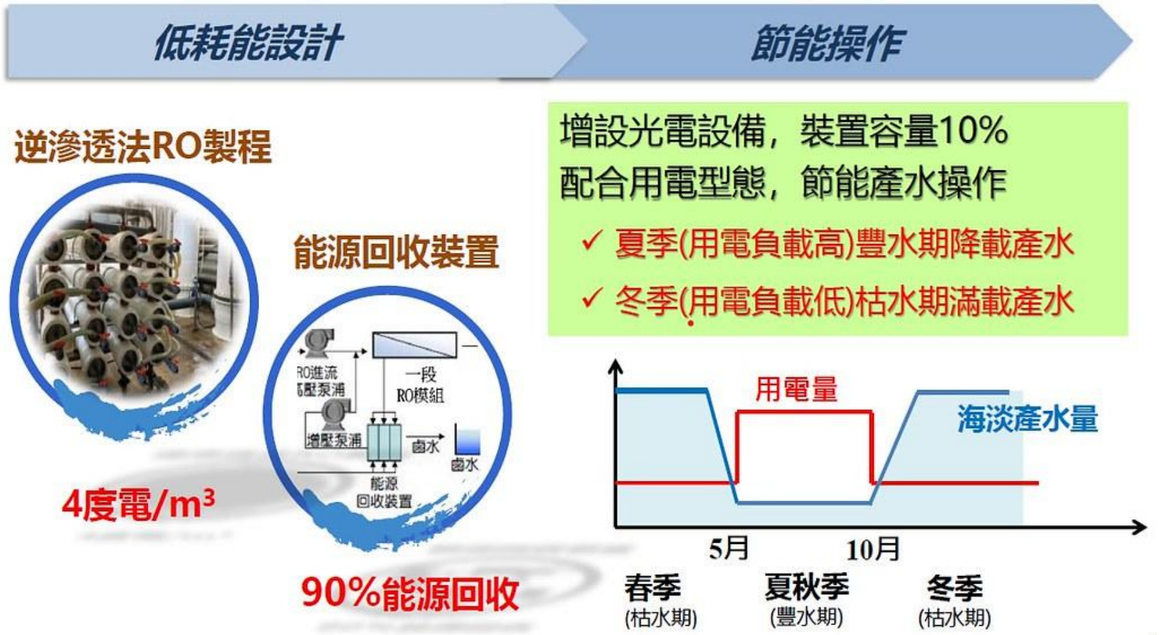
Hsinchu Water Desalination Plant

- Operation begins: 2028
- Capacity: 100,000 cubic meters per day
- To supply industrial parks and for domestic use

Tainan Water Desalination Plants (2-phase)

- Operation begins: after 2028
- Capacity: 200,000 cubic meters per day
- To support technological parks and for domestic use.

Current Issues





Energy Intensive

- Load reduction plans are adopted to reduce energy usage.
- Only 14 hours of water supply capability.

Brine Disposal

- Fishermen’s concerns on impacts of high centration of brine disposal.
- Public doubts the consequences on biodiversity.

This might be Taiwan in 2035, if we don't take actions NOW!

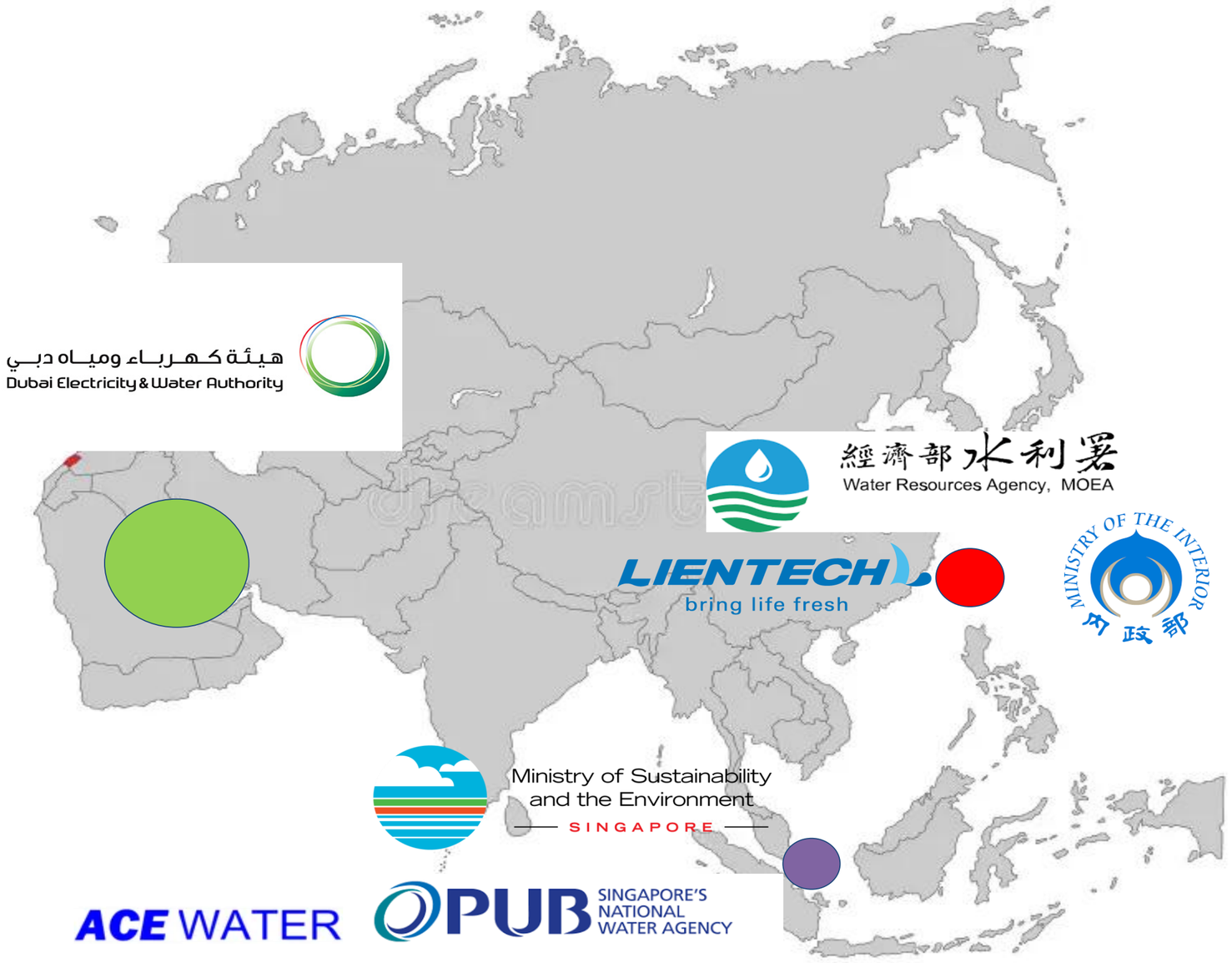








Upcoming Market Expansion



- **Stage 1: Taiwan**

WDP Proportion: *7% (2028)*
*Our solution helps combat increased costs in initial phases.

- **Stage 2: Singapore**

WDP Proportion: 30% (by 2060)

- **Stage 3: Middle East**

WDP Proportion: 70% (by 2040)

Competitor Analysis

Services	SEAsiders	WSP	AECOM	Becht
Cost (million USD)	7	7.5	7.3	8.1
Scalability	✓	✓		✓
Sustainability	✓	✓		
Continuous Quality Control	✓	✓	✓	✓
Financial Planning & AI Modelling	✓			
Energy Recovery and Efficiency	✓		✓	

Testimonials: Proof of Concept



Dr Nurasyikin Binti Misdan

Senior Lecturer and Researcher
Singapore Membrane Technology Centre (SMTC), NTU Singapore | UTHM Malaysia

"This innovative technology can address critical water scarcity by providing an efficient method for desalination, ensuring a reliable supply of fresh water. It generates extensive socio-economic benefits for all involved parties."



Dr Edy Trihatmoko

Researcher
Government National Environment Research and Innovation Agency
Indonesia

"It enhances energy recovery from salinity gradients and offering a renewable energy source which finally reducing the overall power consumption for desalination."



SEASIDERS

Water **A**ccess **T**ransforms **E**very **R**egion

Talent Acquisition Program is LIVE!



- Membrane Technology Specialists
- System Engineers
- Project Managers
- Legal Executives

Ready to make a difference?

Apply NOW to create sustainable water and energy solutions for the future!

Contact Us:

SEASIDERS

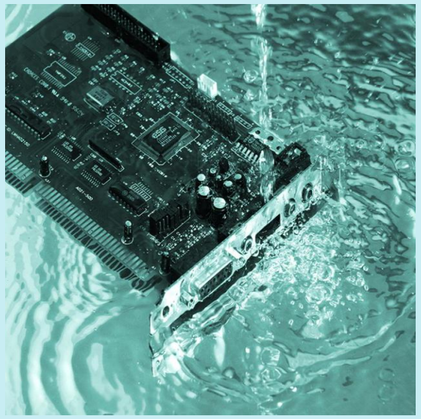
Estimated Financial Projections

No.	Key Components	Details	Estimated Cost (US$)	Estimated Revenue (US$)
1.	Initial Setup Costs	Research and Development	50,000	
		Legal and Administrative	20,000	
		Infrastructure	30,000	
		Marketing and Branding	25,000	
		Human Resources	75,000	
2.	Ongoing Operational Expenses	Salaries and Wages	300,000	
		Office and Utilities	60,000	
		Marketing and Sales	50,000	
		Technology and Maintenance	40,000	
		Miscellaneous	30,000	
3.	Revenue Projections	Consulting Fees		400,000
		Project-Based Revenue		200,000
		Service Packages		100,000
		Partnerships and Collaborations		100,000
Total Estimated Cost			US$680,000	
Total Estimated Return			US$800,000	
Profit			US$120,000	

Clean water changes absolutely **everything**.

If **Taiwan** does not get access to clean water, it affects:

Industrial Needs



Over 40% of semiconductor manufacturers will face high-risk water stress by 2030.
(source: CNA, 2024)

Economy



Taiwan water rates at NT$11 (USD$0.39), which is the lowest in the world, 6 times lower than UK.
(source: Earth.org, 2022)

Access to food



To conserve water, Taiwan has stopped irrigating 74,000 hectares of first crop rice in 2021.
(source: USDA, 2021)

Health



Since April 2024, around 100 village residents had acute gastroenteritis or stomach flu.
(source: CNA, 2024)

Sustainability Impacts





- Improved Water Quality and Efficiency
- Sustainable Water Management and Reduced Pollution





- Enhanced Energy Recovery through Integrated Osmotic Power and Desalination Processes





- 24/7 osmotic power generation supports consistent desalination.
- Clean and renewable energy source for desalination without polluting the air or water.





- Allows for potential cost savings by reducing reliance on external energy sources.

BUSINESS PLAN

Key partners

Research institutions
Utility companies
Manufacturing firms
Technology suppliers
Global distributors
Government agencies
Industry players

Value propositions

Seed funding
Create a R&D team
Do pilot projects and scale-up projects
Regulatory compliance
Full-scale deployment
Global market expansion
Innovation and improvement product's properties

Key resources

R&D team
Pilot project sites
Manufacturing facilities
Expanded capabilities
Regulatory approvals
Brand reputation
Customer support team

Key activities

Short-term:
Innovative osmotic energy solutions
Market research
Public relations
Mid-term:
Supply chain & partnerships
Pre-launch marketing & promotion
Expansion in TW & approach international market
Scalable and efficient energy solutions
Long-term:
Reliable and sustainable energy solutions
Leading-edge technology and market leadership
Expand company size & tailor future roadmap
Expand in global markets
Diversifications
Third party integration

Customer relationships

Collaborative partnerships
Strategic alliances
Service agreement
Contractual relationships
Long-term contracts
Customer services
After-sales support
Maintenance and warranty
Loyalty campaign

Channels

Direct sales
Partnerships
Industry events
Online platform
Trade shows
Global distribution networks
Strategic partnerships

Customer segments

Utilities
Industrial firms
Early adopters
National/ international markets
Multinational corporations
Government entities

Cost structure

Capital Expenditure: Plant construction, site development, equipment installation, permitting, and licensing.
Operational Expenditure: Labor, maintenance, membrane replacement, disposal, insurance, and taxes.
Other Costs: Monitoring and compliance, transportation, and logistics.

Revenue streams

Water Sales: Municipal, industrial, and agricultural supply.
By-Product Sales: Brine and extracted minerals/chemicals.
Government subsidies and grants: subsidies and research grants.
Service Contracts: Maintenance and consulting services.
Technology Licensing: Licensing fees for proprietary technology.

Estimated Cost for 1MW and 10MW

Capital cost



Plant Construction:
$5 million to $20 million



Membrane cost
$1 million to $5 million

Operational and Maintenance Costs



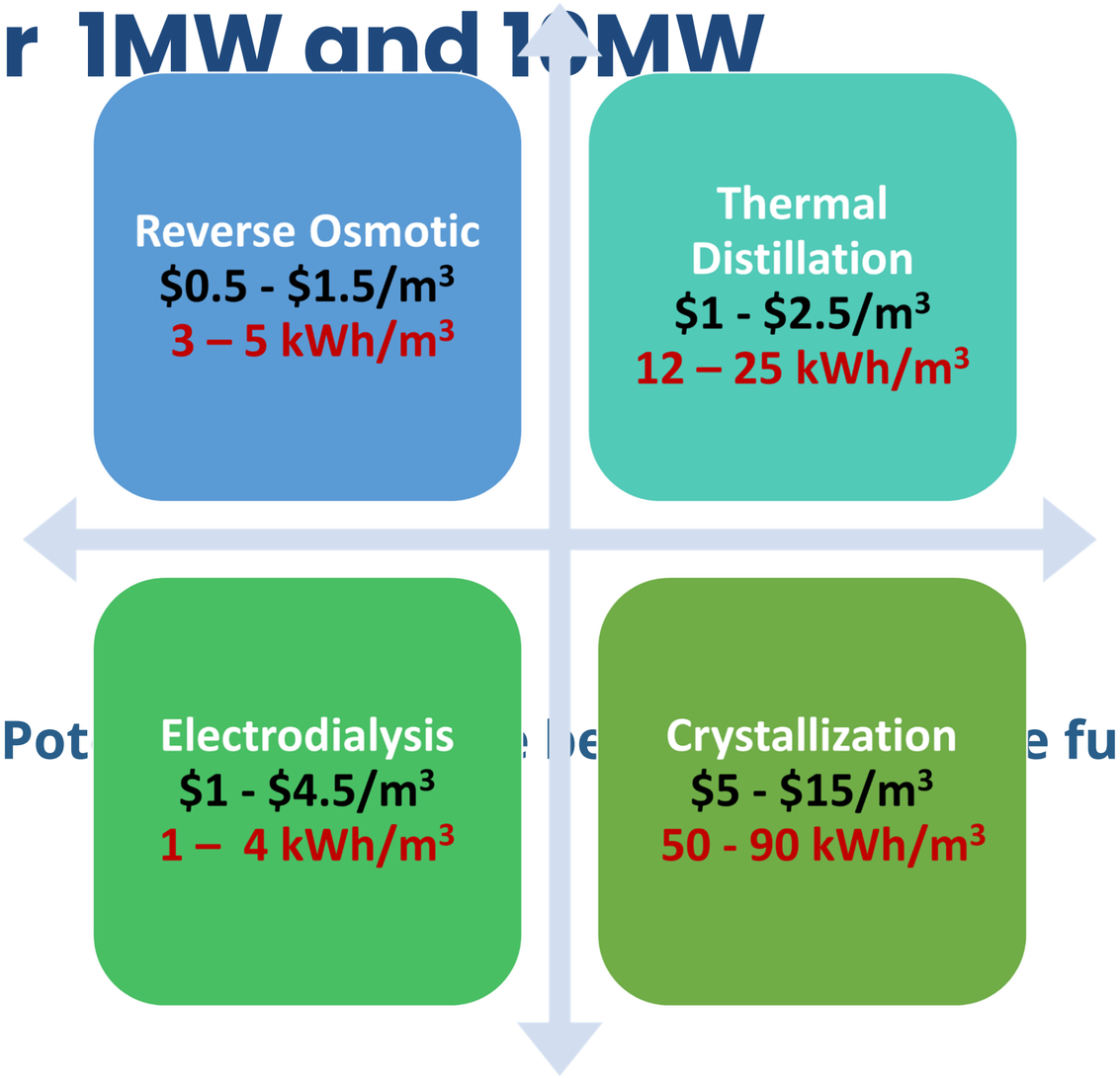
Annual O&M cost
$200,000 to $500,000



Research and Development
$1 million to $5 million

Initial Investment

$7 Million



Pot e future

3 – 5 KWh/m³

EXPECTED MILESTONES WITHIN NEXT 3 YEARS

2024



Start-up Capital

US$2M

- Developing and refining OE technology and solutions.
- Pilot projects to test and validate the technology.

2025



Scaling Up and Testing

US$3M

- Scale up pilot projects to larger installations.
- Collecting performance data and optimizing technology.

2026



Commercialization Preparation

US$6M

- Begin operate commercial osmotic energy solutions.
- Start planning and Obtain necessary regulatory approvals for broader deployment.

US$5M

Receiving revenue from larger pilot projects and initial commercial projects.

2027



Commercialization and Manufacturing

US$10M

- Expand market reach both domestic and international
- Invest money in ongoing R&D to further enhance technology and reduce costs.

US$20M

Our system official launch on the market.

Estimated Total Revenue and Costs for the Next Four Years

Year	Revenue (USD)	Costs (USD)	Breakdown of Costs	Net Profit (Loss) (USD)
2024	$0	$2,500,000	Seed Funding: $1,000,000	-$2,500,000
			R&D: $500,000	
			Pilot Projects: $1,000,000	
2025	$500,000	$3,200,000	Scale-up Pilot Projects: $2,000,000	-$2,700,000
			Data Collection and Optimization: $500,000	
			Regulatory Approvals: $200,000	
			Customer Engagement: $500,000	
2026	$5,000,000	$6,000,000	Commercial-Scale Deployment: $3,000,000	-$1,000,000
			Manufacturing Planning: $1,000,000	
			Operational Optimization: $1,000,000	
			Additional R&D: $1,000,000	
2027	$20,000,000	$10,000,000	Full-Scale Manufacturing: $5,000,000	$10,000,000
			Market Expansion: $2,000,000	
			Ongoing R&D: $1,000,000	
			Operational Costs: $2,000,000	
Net Present Value (NPV)				17,880,916
Internal Return Rate (IRR)				30%

Model		Cost (US$)
1MWh	Operation cost	200
	Retail cost/ 1kWh	0.1 – 0.2
	Retail cost/ 1kLiter	0.4 – 0.6
10 MWh	Operation cost	300
	Retail cost/1kWh	0.05 – 0.1
	Retail cost/1Liter	0.3 – 0.5

Year/Type/Quantity Project		
2024	-	-
2025	Pilot projects	
2026	1 MWh	10
	10 MWh	5
2027	1 MWh	20
	10 MWh	10